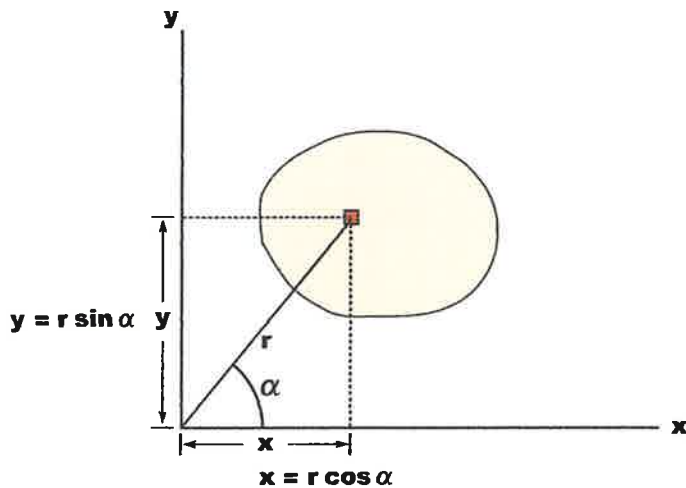


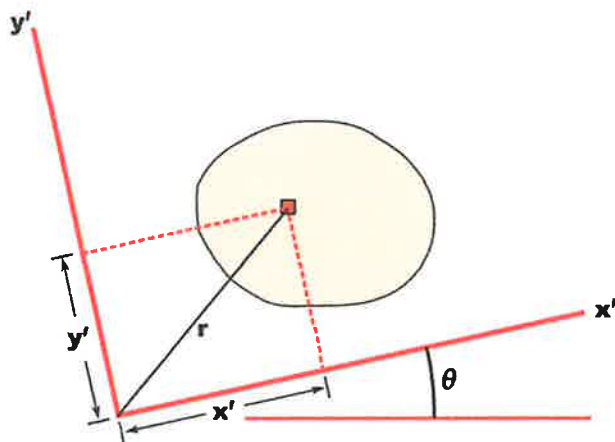
Derivation of Transformation Equations for Area Moments of Inertia

①

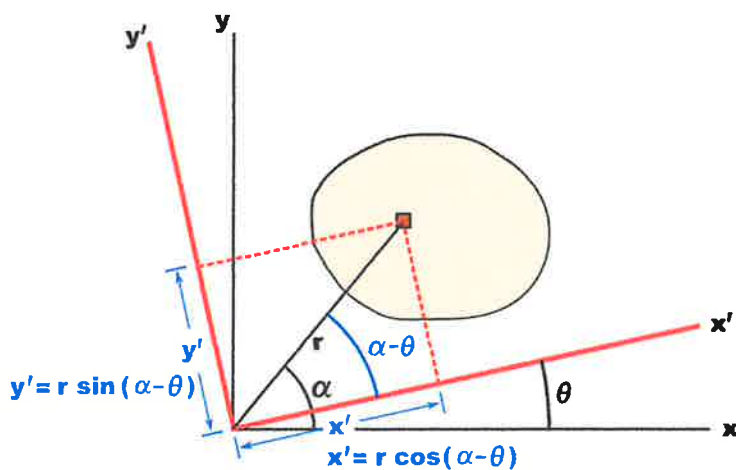


Assume we know
 I_x , I_y and I_{xy} for
 an area at a given
 $y \perp x$ orientation.

Note: $\boxed{\begin{matrix} x = r \cos \alpha \\ y = r \sin \alpha \end{matrix}}$



Goal: To find
 $I_{x'}$, $I_{y'}$ and $I_{x'y'}$
 (can also call these
 I_u , I_v and I_{uv})
 in a rotated coord
 system $y' \perp x'$ from $y \perp x$.
 Rotated by an angle θ .



Writing x' and y'
 in terms of
 r , α , θ .

$\boxed{\begin{matrix} x' = r \cos(\alpha - \theta) \\ y' = r \sin(\alpha - \theta) \end{matrix}}$

In $x'-y'$ coord system (using difference angle identities): ②

$$x' = r \cos(\alpha - \theta) = r (\cos \alpha \cos \theta + \sin \alpha \sin \theta)$$

$$y' = r \sin(\alpha - \theta) = r (\sin \alpha \cos \theta - \cos \alpha \sin \theta)$$

Sub in : $\boxed{x = r \cos \alpha}$; $\boxed{y = r \sin \alpha}$, to get :

$$\boxed{\begin{aligned} x' &= x \cos \theta + y \sin \theta \\ y' &= y \cos \theta - x \sin \theta \end{aligned}}$$

Now calculate the MOI about the x' axis :

$$I_{x'} = \int_A y'^2 dA = \int_A (y \cos \theta - x \sin \theta)^2 dA$$

$$= \int_A (y^2 \cos^2 \theta - 2xy \sin \theta \cos \theta + x^2 \sin^2 \theta) dA$$

$$= \underbrace{\cos^2 \theta \int_A y^2 dA}_{I_x} - 2 \sin \theta \cos \theta \underbrace{\int_A xy dA}_{I_{xy}} + \sin^2 \theta \underbrace{\int_A x^2 dA}_{I_y}$$

$$\textcircled{1} \quad \boxed{I_{x'} = I_x \cos^2 \theta + I_y \sin^2 \theta - 2 I_{xy} \sin \theta \cos \theta}$$

Similarly, for $I_{y'}$

$$I_{y'} = \int_A x'^2 dA = \int_A (x \cos \theta + y \sin \theta)^2 dA$$

$$\textcircled{2} \quad \boxed{I_{y'} = I_x \sin^2 \theta + I_y \cos^2 \theta + 2 I_{xy} \sin \theta \cos \theta}$$

For $I_{x'y'}$

③

$$\begin{aligned} I_{x'y'} &= \int_A x'y' dA = \int_A (x \cos \theta + y \sin \theta) \times \\ &\quad (y \cos \theta - x \sin \theta) dA \\ &= \int_A (xy \cos^2 \theta - xy \sin^2 \theta - x^2 \sin \theta \cos \theta + y^2 \sin \theta \cos \theta) dA \end{aligned}$$

$$\textcircled{3} \quad \boxed{I_{x'y'} = I_{xy} (\cos^2 \theta - \sin^2 \theta) + (I_x - I_y) \sin \theta \cos \theta}$$

I call these (①, ②, ③) the " θ " forms of the transformation equations.

Note: If add ① + ②, you get:

$$\begin{aligned} I_{x'} + I_{y'} &= I_x (\cos^2 \theta + \sin^2 \theta) \\ &\quad + I_y (\sin^2 \theta + \cos^2 \theta) + 0 \end{aligned}$$

"1" "1"

Finally:

$$\boxed{I_{x'} + I_{y'} = I_x + I_y} \quad \boxed{\begin{aligned} &= I_{\min} + I_{\max} \\ &= I_u + I_v \end{aligned}}$$

This is a useful result.

Alternate forms of the transformation eqns: ⁽⁴⁾
(I call these the "2 θ " forms)

Use these trig identities:

$$2 \sin \theta \cos \theta = \sin 2\theta$$

$$\cos^2 \theta - \sin^2 \theta = \cos 2\theta$$

$$\cos^2 \theta = \frac{1 + \cos 2\theta}{2} ; \quad \sin^2 \theta = \frac{1 - \cos 2\theta}{2}$$

$$I_{x'} = I_x \cos^2 \theta + I_y \sin^2 \theta - 2 I_{xy} \sin \theta \cos \theta$$

$$= I_x \left(\frac{1 + \cos 2\theta}{2} \right) + I_y \left(\frac{1 - \cos 2\theta}{2} \right) - I_{xy} \sin 2\theta$$

$$\textcircled{4} \quad I_{x'} = \left(\frac{I_x + I_y}{2} \right) + \left(\frac{I_x - I_y}{2} \right) \cos 2\theta - I_{xy} \sin 2\theta$$

$$I_{y'} = I_x \sin^2 \theta + I_y \cos^2 \theta + 2 I_{xy} \sin \theta \cos \theta$$

$$= I_x \left(\frac{1 - \cos 2\theta}{2} \right) + I_y \left(\frac{1 + \cos 2\theta}{2} \right) + I_{xy} \sin 2\theta$$

$$\textcircled{5} \quad I_{y'} = \left(\frac{I_x + I_y}{2} \right) - \left(\frac{I_x - I_y}{2} \right) \cos 2\theta + I_{xy} \sin 2\theta$$

$$I_{x'y'} = I_{xy} (\cos^2 \theta - \sin^2 \theta) + (I_x - I_y) \sin \theta \cos \theta$$

$$\textcircled{6} \quad I_{x'y'} = I_{xy} \cos 2\theta + \left(\frac{I_x - I_y}{2} \right) \sin 2\theta$$

Note: If add $\textcircled{4} + \textcircled{5}$, get:

$$I_{x'} + I_{y'} = \left(\frac{I_x + I_y}{2} \right) + \left(\frac{I_x + I_y}{2} \right) + 0 + 0$$

$$\boxed{I_{x'} + I_{y'} = I_x + I_y} \quad \boxed{= I_{\min} + I_{\max}} \\ \boxed{= I_u + I_v}$$

So, these $I_{x'}$, $I_{y'}$ and $I_{x'y'}$ equations ⑤
(can also call them I_u , I_v , I_{uv}) equations
give values of I_u , I_v , I_{uv} in a coord system
rotated an angle θ from the orig x - y system.

"Principal" values of $I_{x'}$, $I_{y'}$ are also
called I_{MAX} and I_{MIN} .

These occur at angles θ_{p1} and θ_{p2} ,
which are called "principal angles."

How find principal values I_{MIN} and I_{MAX} ?

How find principal angles?

One way: Start w/ the $I_{x'}$ eqn (the 2θ form)

$$I_{x'} = \left(\frac{I_x + I_y}{2} \right) + \left(\frac{I_x - I_y}{2} \right) \cos 2\theta - I_{xy} \sin 2\theta$$

Take its derivative and set $= 0$, to find max
and min:

$$\frac{dI_{x'}}{d\theta} = 0 + \left(\frac{I_x - I_y}{2} \right) (-2 \sin \theta) - I_{xy} (2 \cos 2\theta) = 0$$

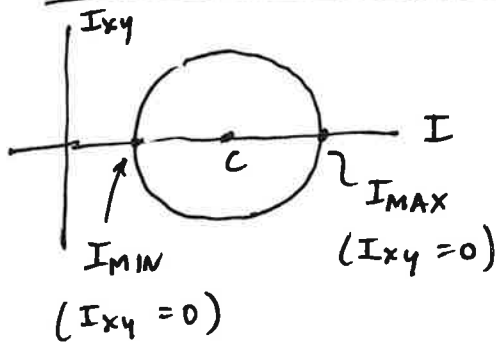
We get;

$$\boxed{\tan 2\theta_p = \frac{-I_{xy}}{\left(\frac{I_x - I_y}{2} \right)}}$$

Knowing I_x , I_y , I_{xy} , this gives us θ_p ,
which is the angle to the nearest principal
value, I_{MIN} or I_{MAX} .

Alternate way of getting θ_p .

⑥



One thing you'll learn about Mohr's Circle is that at I_{MAX} and I_{MIN} , $I_{xy} = 0$.

So, you can use the $I_{x'y'}$ eqn:

$$I_{x'y'} = I_{xy} \cos 2\theta + \left(\frac{I_x - I_y}{2} \right) \sin 2\theta = 0$$

set it = 0 ... Solve for θ_p :

$$\tan 2\theta_p = \frac{-I_{xy}}{\left(\frac{I_x - I_y}{2} \right)}$$

Once have θ_p , then can plug this into the $I_{x'}$ and $I_{y'}$ equations to get I_{MIN} , I_{MAX} .

But to me, it's easier to set up Mohr's Circle to get θ_p 's, I_{MIN} , I_{MAX} , and visualizing the rotation needed to go from I_x , I_y , I_{xy} to I_{MAX} and I_{MIN} .